

Problem Solving (MATH 1000)

Summer 2018

Thursday, 05/24/2018

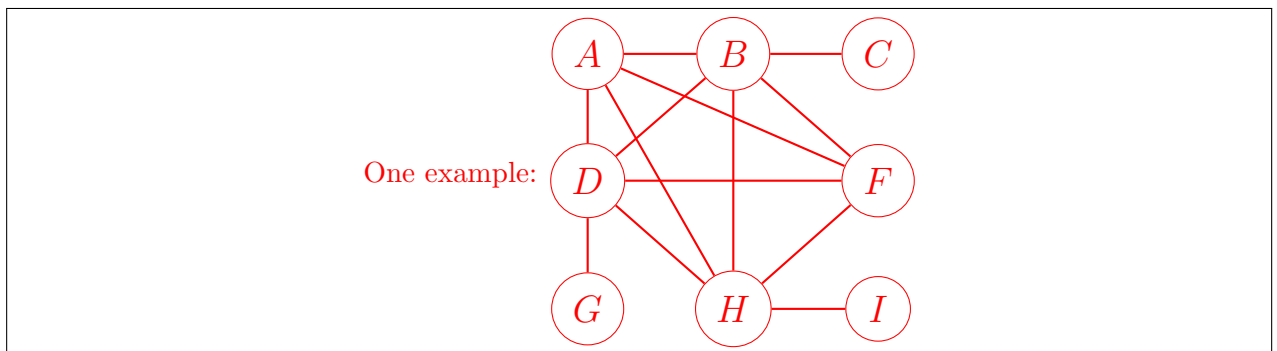
Quiz 1: Section 5.2, 5.3, 6.2, 6.4

Name (Print): _____

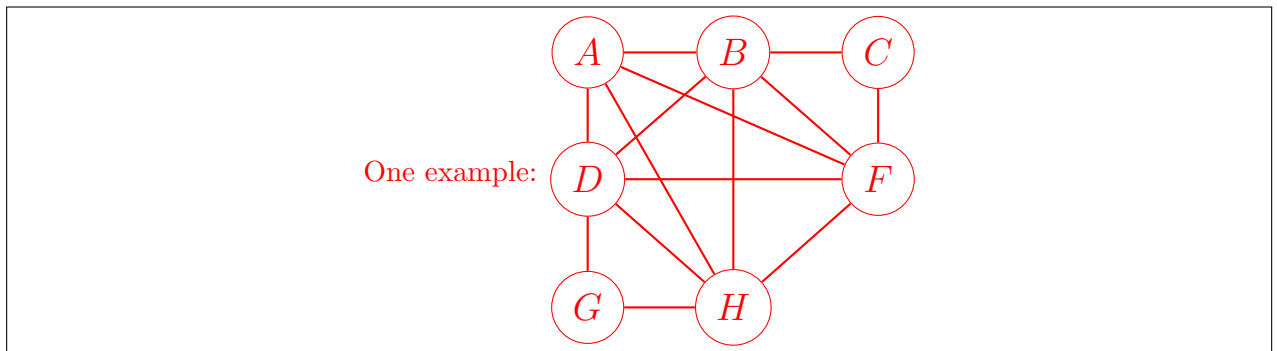
Score: _____ / 100

1. For each of the following items below, draw an example of a graph that satisfies the given properties:

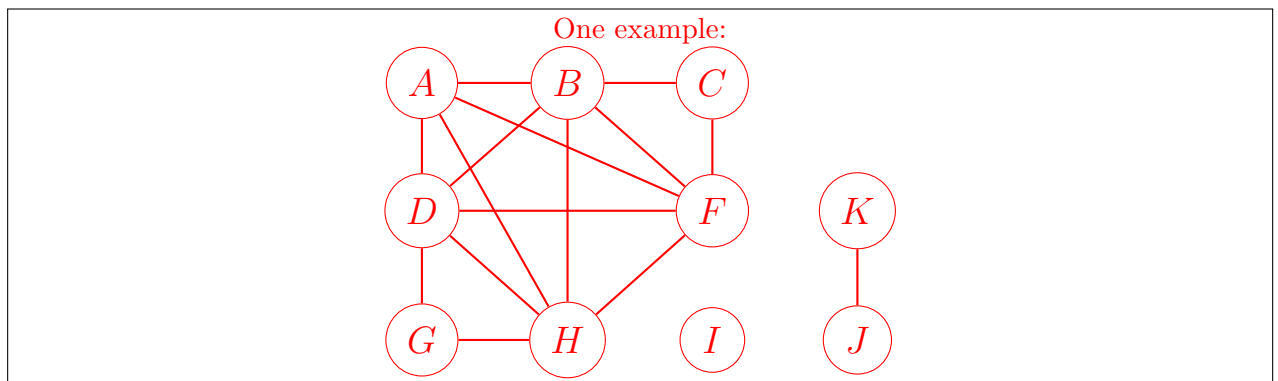
- (a) 10 points A graph with exactly 8 vertices and exactly 3 bridges.



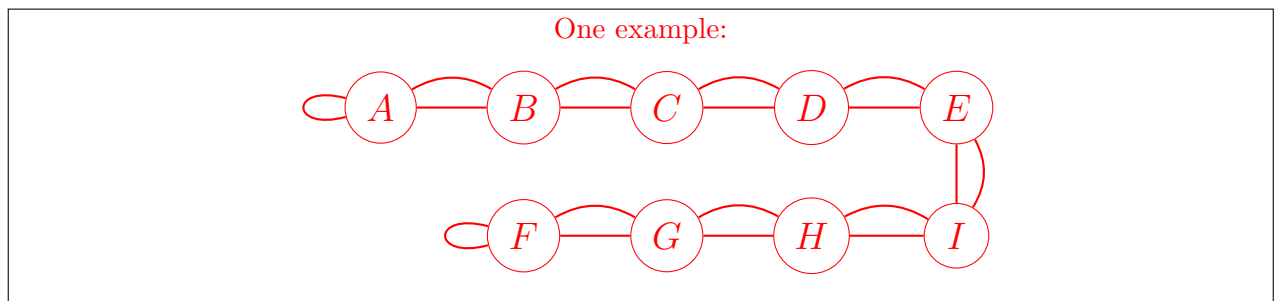
- (b) 10 points A graph with exactly 7 vertices and no bridges.



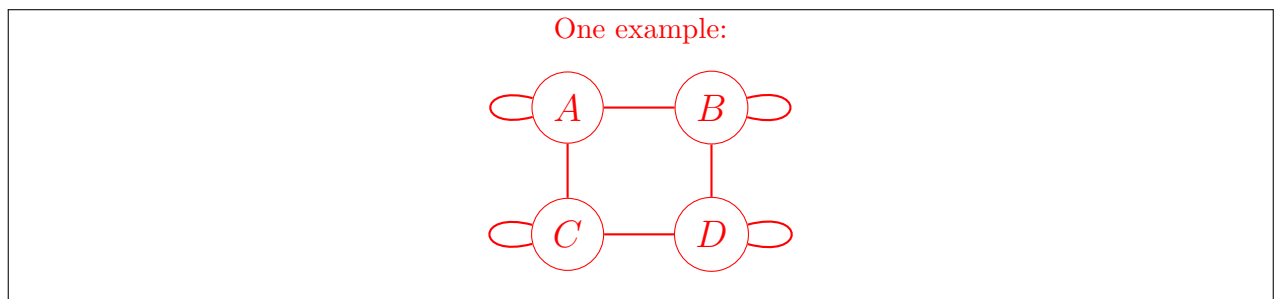
- (c) 10 points A graph with at least 10 vertices, at least 9 edges, and exactly 3 components.



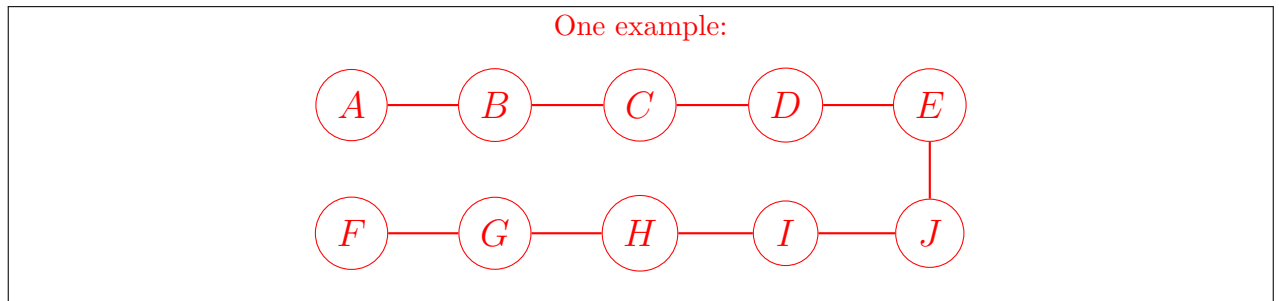
2. 10 points Draw an example of a connected graph with exactly 9 vertices that is not simple but is regular and has two vertices with five degrees of separation.



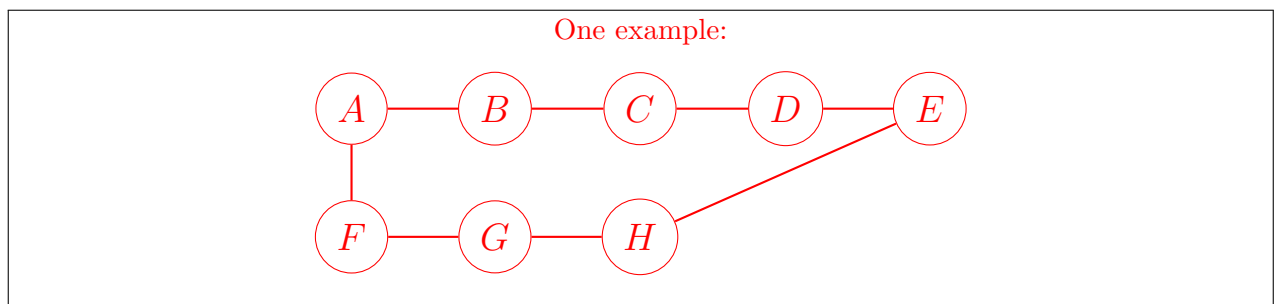
3. 10 points Draw an example of a graph with up to 4 vertices that is not simple but is regular.



4. 10 points Draw a graph that has exactly 10 vertices, and has an Euler Path.



5. 10 points Draw a graph that has exactly 8 vertices, and has an Euler Circuit.



6. 10 points State Euler's sum of degrees theorem and briefly explain why this theorem is true (you may be casual with your phrasing in the explanation – I just want to see that you understand the concept on an intuitive level).

Euler's sum of degrees theorem states that the sum of the degrees of all the vertices in a graph is equal to twice the number of edges in the graph. This follows from the fact that each edge contributes two degrees – that is, it joins two vertices and increases each vertex's degree by one (or in a loop, joins a vertex to itself and increases its degree by two).

7. 10 points Draw any graph which has a Hamilton Circuit but not a Hamilton Path. However, if this is impossible, clearly and succinctly explain why.

This is impossible – any graph with a Hamilton circuit automatically has a Hamilton path. This follows from the fact that we may construct a Hamilton path in a graph with a Hamilton circuit by removing a single edge from the circuit (it doesn't matter which one, since it is a circuit).

8. 10 points Draw a connected, simple graph with six vertices which has no Euler Circuit, no Euler Path, no Hamilton Circuit, and no Hamilton Path. However, if this is impossible, clearly and succinctly explain why.

One example:

